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“An infrastructure optimised for voice was an expensive way to transmit data; we had to change it”

Tejas Networks designs end-to-end data and telecom solutions based on optical networking technology. Arnob Roy, president-engineering, Tejas Networks, spoke to Abhishek Mutha of EFY about the seamless migration from 3G to 4G networks, the challenges faced and the exciting trends in the telecom sector

What kind of technologies are used by engineers in the optical communications field?

A company into optical communications builds transmission systems for voice, data and video based on optical fibre, and designs electronics that forms the data transmission backbone. In terms of specific technologies, considering the voice networks, some of the prominent technologies worked with are synchronous optical networking (SONET), synchronous digital hierarchy (SDH) and optical transport network (OTN). For data transmission, we have Carrier Ethernet technologies like multi-protocol label switching transport layer and Ethernet ring protection switching.

Within the communication products portfolio, we develop and design high-speed telecom systems. This requires multidisciplinary skills. We have high-speed PCBs, FPGAs (the silicon component that goes into designing the product), huge amount of embedded software that runs on this product and network management software (the software that runs outside the product managing larger networks).

What is the most exciting technology in optical communications at the moment?

Micro packet optical transport is one of the technologies that we have been working on for the last few years. At the moment, we are working on next-generation transmission technologies based on packet transport network, and efficient energy management solutions for telecom tower operators.

What are the challenges faced in transitioning from 2G to 3G/4G, and how do your solutions tackle them?

Initially, for a 2G network, sending data over time-

division multiplexing (TDM) network is the conventional way. When the traffic increases, it is better to send data over a separate packet network rather than over the TDM network.

The voice infrastructure is supported in the usual way, i.e., TDM, but sending data over a separate packet data network rather than over a voice network is far more cost-efficient as well as bandwidth-efficient. When 4G will come and when the dominant traffic in the network would be packets/data, voice transmission is going to be more efficient over the data network by packetising it.

Our product allows the network to transition seamlessly from one network architecture to another without having to change your hardware. It is basically the reprogramming of your hardware that allows you to do that.

We achieved this through a lot of programmable silicon like FPGAs and by reprogramming the data path to move from TDM to packet and a combination of both, and so on.

What makes transition from one communication technology to the next so cumbersome?

The development of packet optical system architecture has itself been a really interesting challenge. This is a system that helps in transition from a 2G to 3G to 4G network, which requires an enormous amount of flexibility in the product in terms of hardware as well as programmability.

A few years ago, the conventional network was mainly voice and the efficient way of transmitting voice traffic was through time-division multiplexing (TDM) technologies like SONET and SDH. When people started accessing data